

Saved-Model Evaluation and Analysis: Five Bruise-Segmentation Models on the 28-Subject Consensus Test Set

Injury Analytics Lab — Bruise Segmentation Project

Generated from `bruise_colab_saved_eval.ipynb` and
`bruise_colab_saved_analysis.ipynb`
(the project’s original saved checkpoints; results in `results_final/*saved*.zip`)

Abstract

This report documents the evaluation and full visual analysis of the project’s five *saved* segmentation models—a SegFormer-B2 teacher, SegFormer-B0 direct and distilled students, and two YOLO26n-sem models—loaded as trained checkpoints (no retraining) and scored on the fixed **185-image / 28-subject** test set against the three-annotator **2-of-3 majority** target. Each model’s decision threshold is fitted on the 134-image validation set with a 1-D logit sweep (median of the 1-SE tie band). The SegFormer models cluster at 0.773–0.786 mean Dice with a **0–0.5% complete-miss rate**; here the **distilled B0 (0.786) is the top scorer**, edging the B2 teacher (0.773). The YOLO models reach a comparable median (~ 0.85 native) but a much lower mean (0.69–0.74) and a **7–10% miss rate**. As with every analysis in this project: (i) model-vs-model Dice differences are not resolvable at $n = 28$ (distillation +0.009, student–teacher +0.013, SegFormer–YOLO +0.051 all n.s.); (ii) the resolvable axis is the complete-miss rate; (iii) the three human experts agree with one another at only **0.639** mean Dice and every model beats that ceiling—indeed all three SegFormer models match the consensus *significantly better than the annotator (Paul) they were trained on* (+0.07–0.09, $P \geq 0.99$); and (iv) an apparent light-skin deficit is partly a lesion-size artefact but a residual light-skin *recall* deficit survives size and subject clustering for most models. Twenty-two figures are included.

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1 Scope and Setup

These are the five *already-trained* models saved in the project’s local directories (the checkpoints used across prior presentations), loaded and scored **without any retraining**. The evaluation protocol is identical to the rest of the project:

- **Test set:** 185 white-light images from 28 subjects; target = per-pixel 2-of-3 majority vote of three experts (Paul, Gbarimah, Erik). Scored on a 640×640 grid.
- **Threshold:** fitted on the 134-image *validation* set only, by sweeping a 1-D cut on the raw bruise logit and choosing the median of all cuts within one standard error of the peak (a stable “tie-band” rule; Figure 1).
- **YOLO** is scored two ways: *native argmax* (its home-turf letterbox path) and *custom /255* (the same 640 geometry SegFormer sees). Unless noted, the analysis uses native argmax as the core YOLO number.
- **Statistics:** the 185 images are only 28 subjects, so all intervals are **subject-level cluster bootstraps** (resample subjects, $B = 4000$); model-vs-model contrasts are *paired*. *These are single saved checkpoints, so there is no training-seed spread—the intervals reflect test-set (subject) sampling only.*

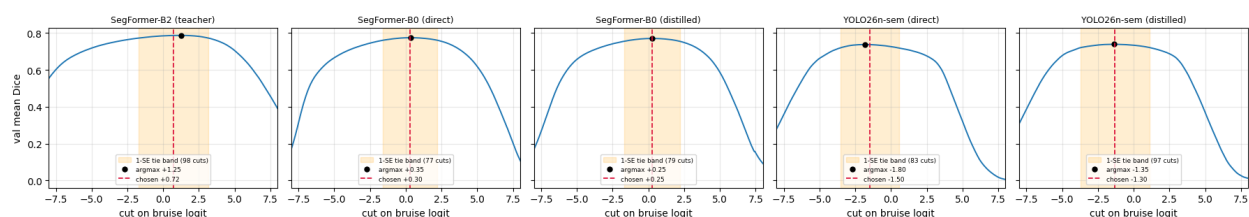


Figure 1: Validation threshold sweep for each model (mean Dice vs. the logit cut). The flat top is the point: over a wide band of cuts the threshold is underdetermined, so we take the median of the 1-SE tie band (dashed line) rather than the noisy argmax (dot).

2 Accuracy and Distribution

Table 1 is the master result table (from `saved_eval`); the YOLO rows there are the custom-/255 path. Figure 2 gives the headline Dice (YOLO shown at native argmax, its stronger median).

Table 1: Master results: threshold fitted on val, scored on test (185 images). YOLO rows are the custom-/255 path. Latency is single-frame 640→640 on the GPU.

Model	Params	Thr.	Median Dice	Mean Dice	Miss %	Median ms	FPS
SegFormer-B2 teacher	27.35M	0.674	0.834	0.773	0.00	32.3	30.8
SegFormer-B0 direct	3.71M	0.574	0.813	0.777	0.54	15.9	62.6
SegFormer-B0 distilled	3.71M	0.562	0.828	0.786	0.54	15.9	62.6
YOLO26n direct (custom)	1.63M	0.182	0.825	0.699	9.19	7.9	125.2
YOLO26n distilled (custom)	1.63M	0.214	0.805	0.692	5.95	7.9	126.8

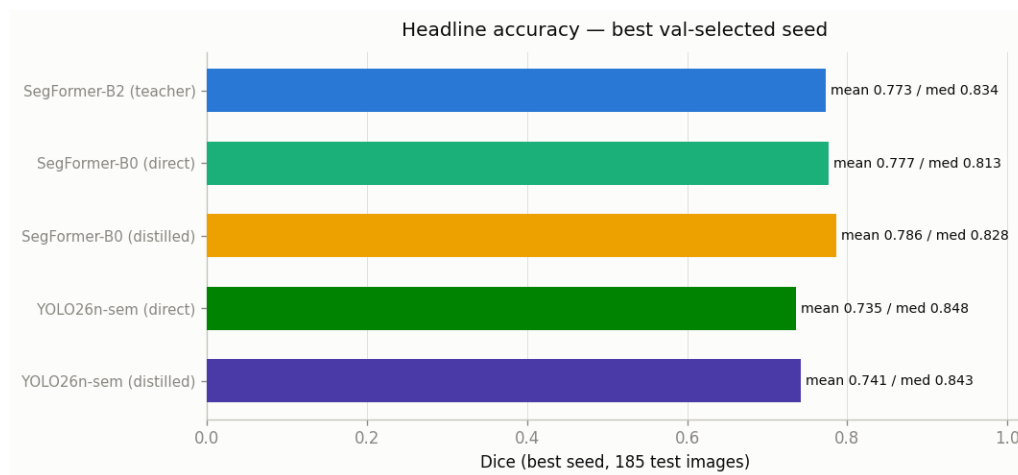


Figure 2: Headline Dice (YOLO at native argmax). The **distilled B0 is the top mean scorer (0.786)**, ahead of the B2 teacher (0.773)—i.e. on these saved checkpoints the “student beats teacher” pattern holds (it did *not* survive the batch-equalised final retraining, but it does here).

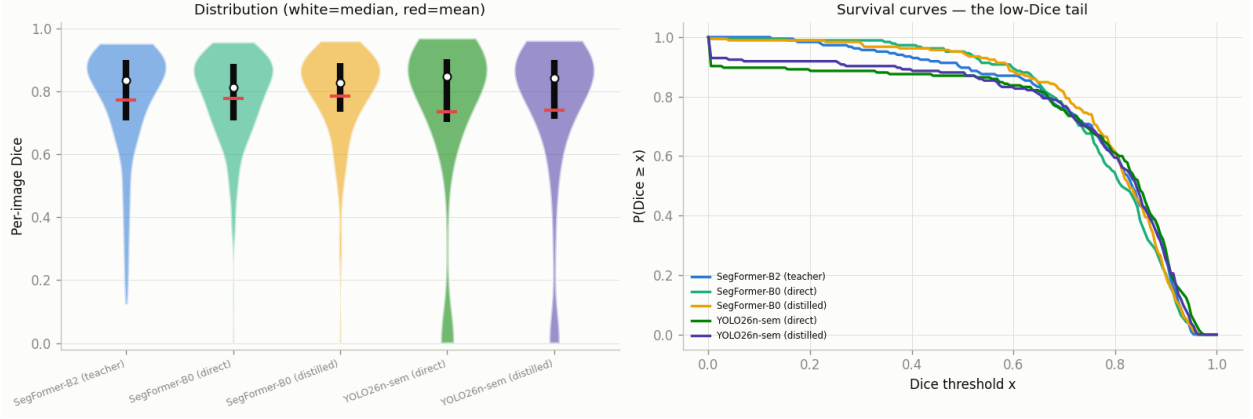


Figure 3: Left: per-image Dice distribution (white = median, red = mean); the long low-Dice tail is why we report the median and treat failures separately. Right: survival curves $P(\text{Dice} \geq x)$ —the YOLO curves drop earliest at low x , i.e. more near-zero images.

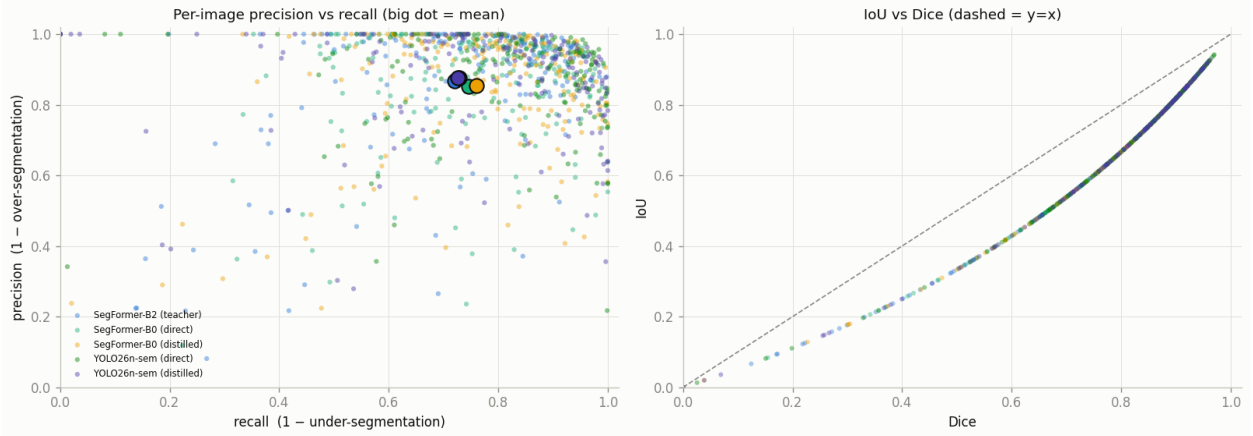


Figure 4: Left: per-image precision vs. recall (big dot = mean). Right: IoU vs. Dice—a monotone consistency check. YOLO tends to sit high-precision / lower-recall (clean but under-firing), which foreshadows its miss and fairness behaviour.

3 Safety and Failure Modes

The decisive failure for an injury-documentation tool is a **complete miss**: a real bruise, empty mask. Figure 5 shows the miss rates and the “best Dice $\neq$ safest model” effect—the YOLO models have the highest *median* Dice yet by far the highest miss rate.

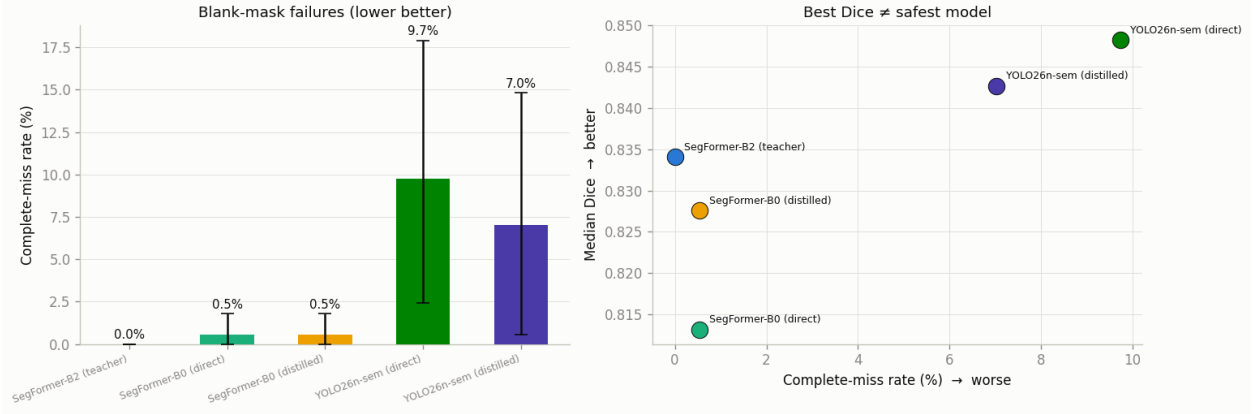


Figure 5: Complete-miss rate with subject-bootstrap intervals (left) and miss vs. median Dice (right). SegFormer-B2 never blanks (0%); the YOLO models blank 7–10% of images despite the best medians.

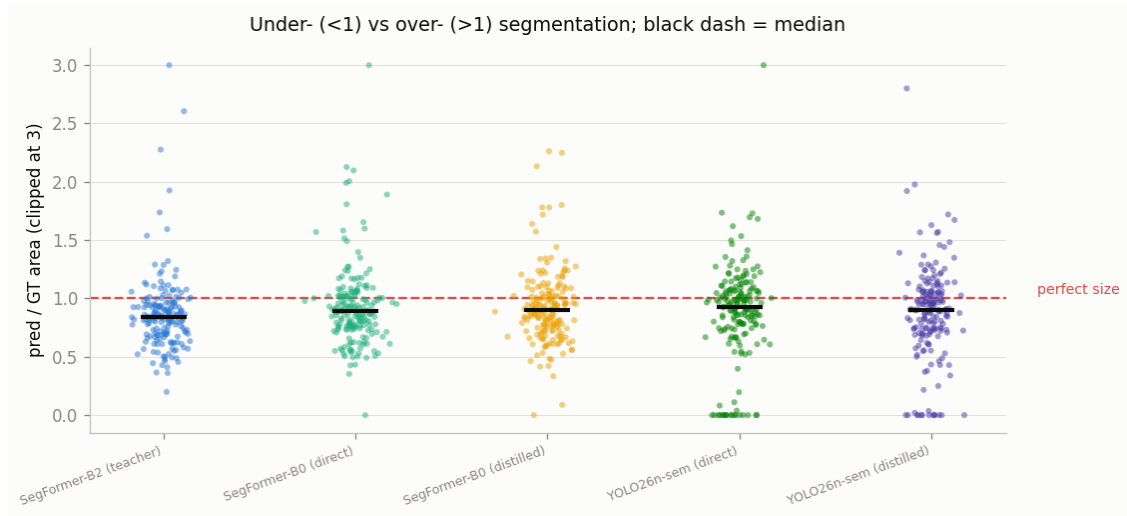


Figure 6: Over- vs. under-segmentation: predicted÷true bruise area (black dash = median). All models sit slightly below 1 (mild under-segmentation); the YOLO models carry a spike of images at ratio 0 (the blank masks).

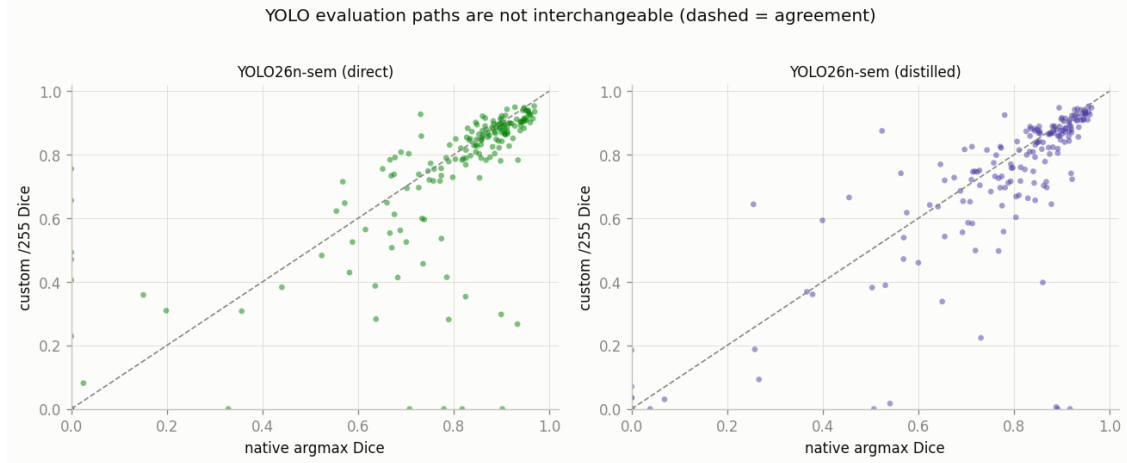


Figure 7: YOLO’s two evaluation paths are not interchangeable: native argmax vs. custom /255 per image. Same weights, different eval—report both and state which.

4 Which Differences Are Real?

Every marginal interval (Figure 8) overlaps heavily; the paired contrasts (Table 2, Figure 9) confirm that **only the miss-rate gap clears zero**. Notably, on these saved checkpoints even SegFormer–YOLO on Dice is *not* significant (+0.051, CI spans zero), unlike the batch-equalised final run.

Table 2: Paired subject-level cluster-bootstrap contrasts ($B = 4000$). Dice rows except the last (complete-miss rate, percentage points).

Contrast	Δ	95% CI	$P(\Delta > 0)$	Verdict
Distillation (B0-dist – B0-direct)	+0.009	[−0.009, +0.029]	0.82	n.s.
Student – teacher (B0-dist – B2)	+0.013	[−0.023, +0.052]	0.74	n.s.
SegFormer – YOLO (B0-dist – YOLO-direct)	+0.051	[−0.011, +0.117]	0.94	n.s.
YOLO distilled – direct	+0.006	[−0.019, +0.030]	0.67	n.s.
MISS%: YOLO-direct – B0-dist	+9.19	[+1.83, +17.54]	1.00	sig.

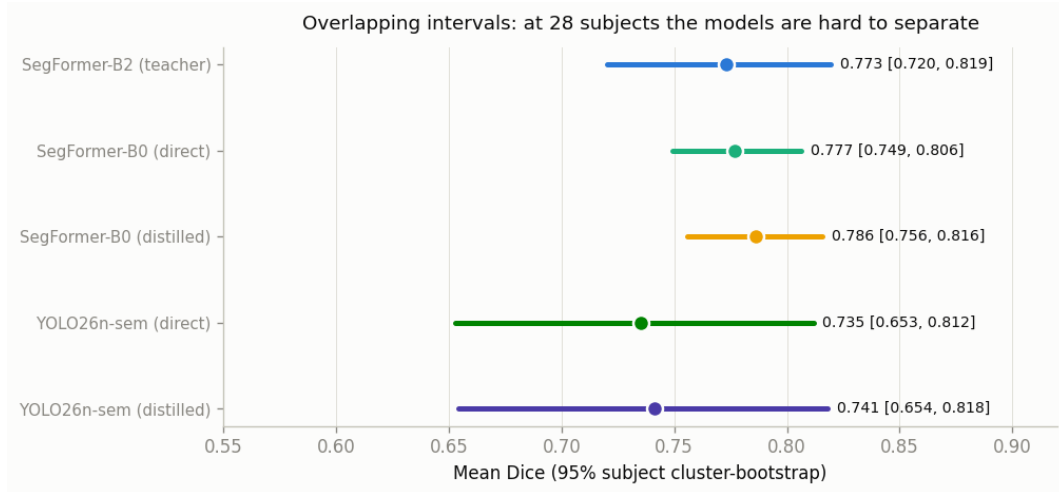


Figure 8: Marginal mean-Dice intervals. At 28 subjects the five models are hard to separate.

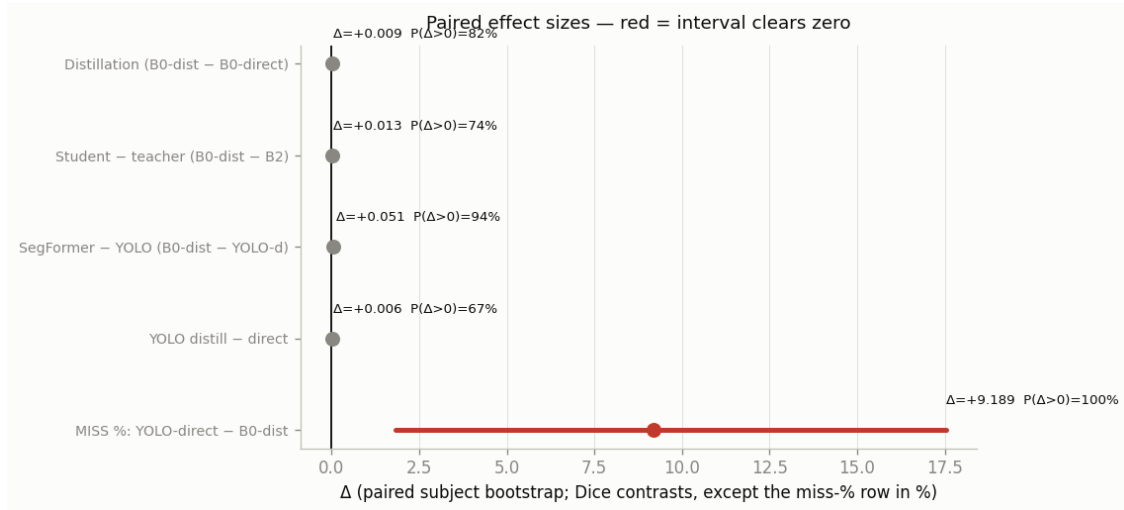


Figure 9: Paired effect sizes; red = interval clears zero. Only the miss-rate contrast is resolvable. $P(\Delta > 0)$ distinguishes an underpowered null (distillation 82%) from a coin flip (student-teacher 74%).

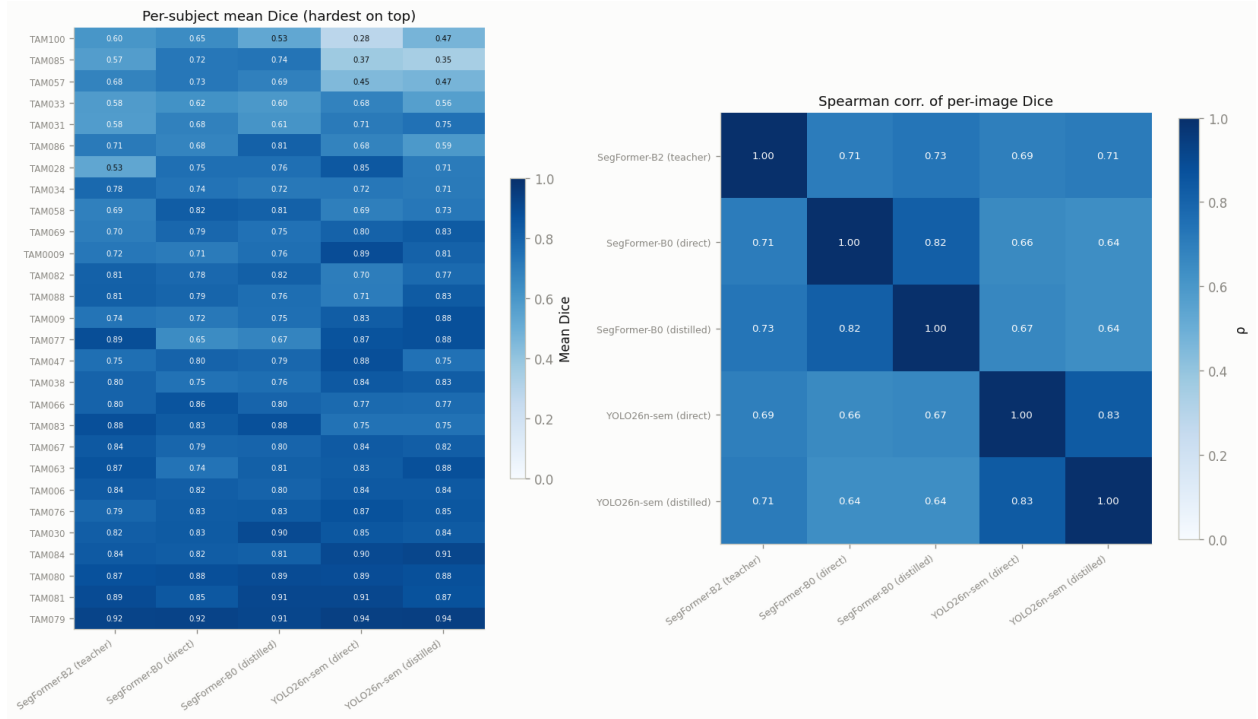


Figure 10: Left: per-subject mean Dice (hardest subjects on top); rows dark across every column are hard *subjects*, not model failures—and with only 28 rows, one or two of them move the whole average. Right: Spearman correlation of per-image Dice; the models largely agree on which images are hard.

5 Fairness Across Skin Tone (ITA)

Groups are ITA (objective, pixel-computed). This is **exploratory** at $n = 28$ (≈ 9 – 17 subjects/group). SegFormer-B2 and B0-direct show a nominally significant omnibus effect (Kruskal $p = 0.016, 0.027$); the striking pattern is YOLO’s **light-skin under-detection**: its complete-miss rate on light skin reaches **28%** (native direct) against 0–7% elsewhere (Figure 11, right).

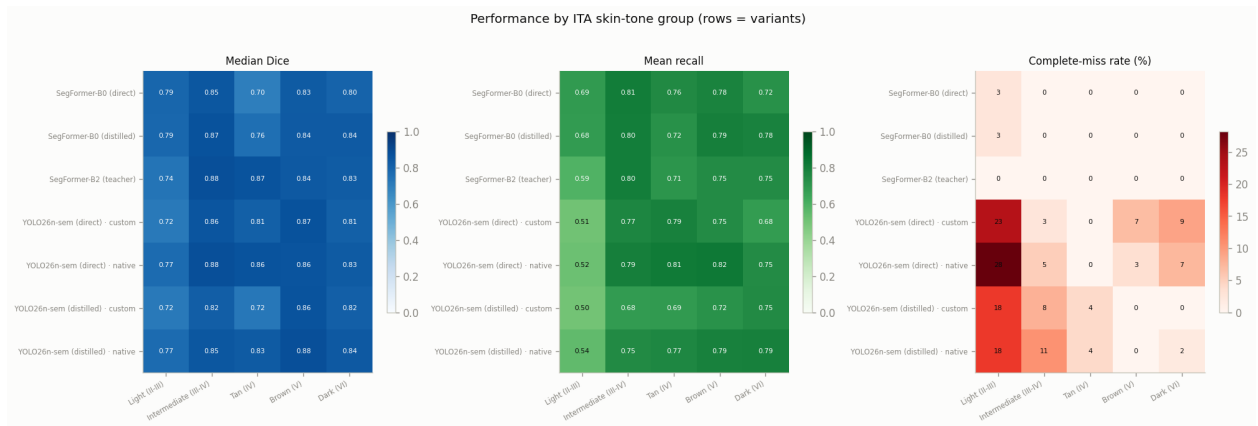


Figure 11: Per-ITA-group median Dice (left), mean recall (middle), and complete-miss rate (right) for every variant. The miss panel is the story: the YOLO variants concentrate their blank masks on the lightest-skin group.

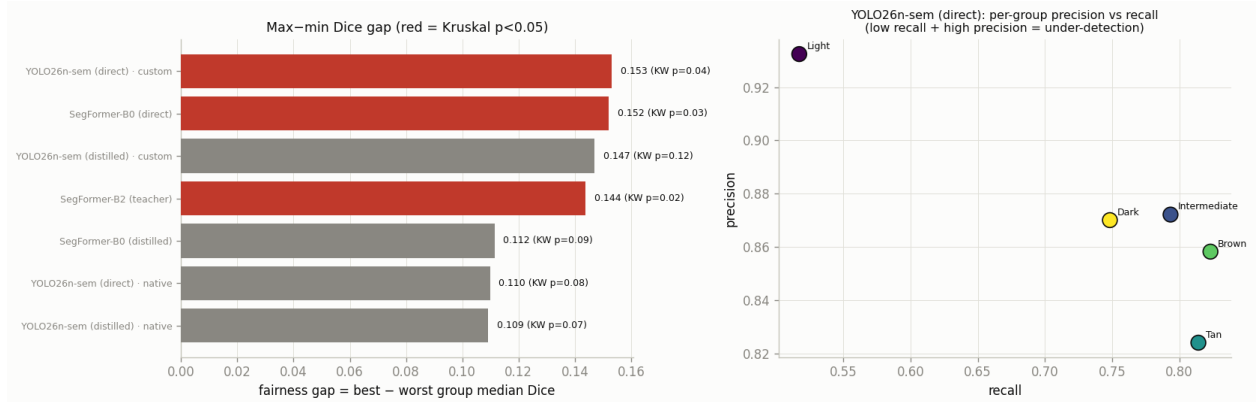


Figure 12: Left: best–worst-group median-Dice gap per variant (red = Kruskal $p < 0.05$). Right: YOLO-direct per-group precision vs. recall—light skin sits at high precision / very low recall, i.e. *under-detection*, not confusion.

5.1 Is the light-skin gap just lesion size?

Light-skin bruises are the *smallest* in this cohort (Figure 13), and small bruises are harder for every model—so size is a confound. We test whether the signal survives it with subject-clustered regressions (Table 3) and stratification (Figure 14). **It largely survives:** the light coefficient on recall stays significant with $\log(\text{size})$ controlled for the B2 teacher (-0.195 , $p = 0.003$), B0-direct (-0.098 , $p = 0.016$), and both YOLO models ($-0.255/-0.246$, $p = 0.014/0.003$); B0-distilled is borderline ($p = 0.057$). Within the smallest size tercile, YOLO-direct still blanks **35%** of light-skin images vs. 9.5% of the rest. This is a stronger residual signal than the final batch-equalised run showed—still a hypothesis at 9 light-skin subjects, but a consistent one.

Table 3: Light-skin coefficient with $\log_{10}(\text{size})$ controlled, subject-clustered SEs. Negative = worse on light skin at equal size.

Model	Outcome	light coef	p (clustered)
SegFormer-B2 teacher	recall	-0.195	0.003
SegFormer-B0 direct	recall	-0.098	0.016
SegFormer-B0 distilled	recall	-0.097	0.057
YOLO26n direct	recall	-0.255	0.014
YOLO26n distilled	recall	-0.246	0.003
YOLO26n direct	miss (logit)	$+1.56$	0.088

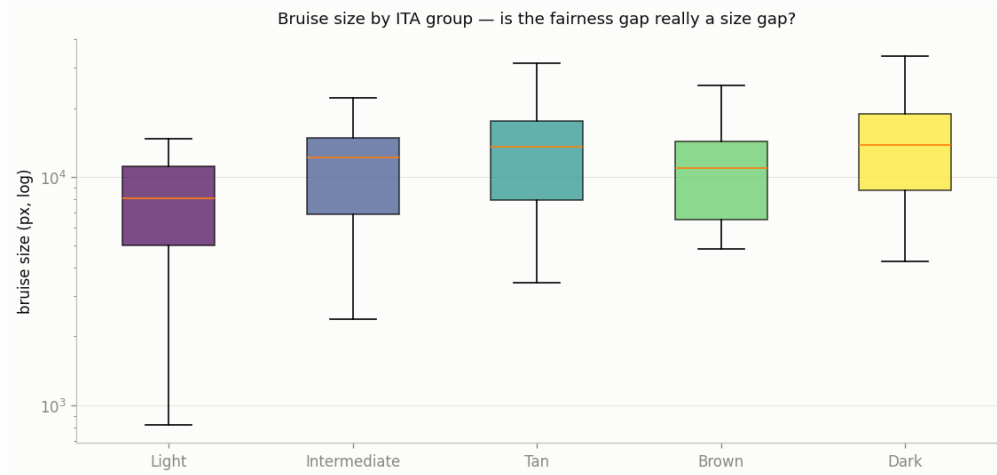


Figure 13: Bruise size by ITA group. Light skin has the smallest lesions—the confound that Table 3 and Figure 14 control for.

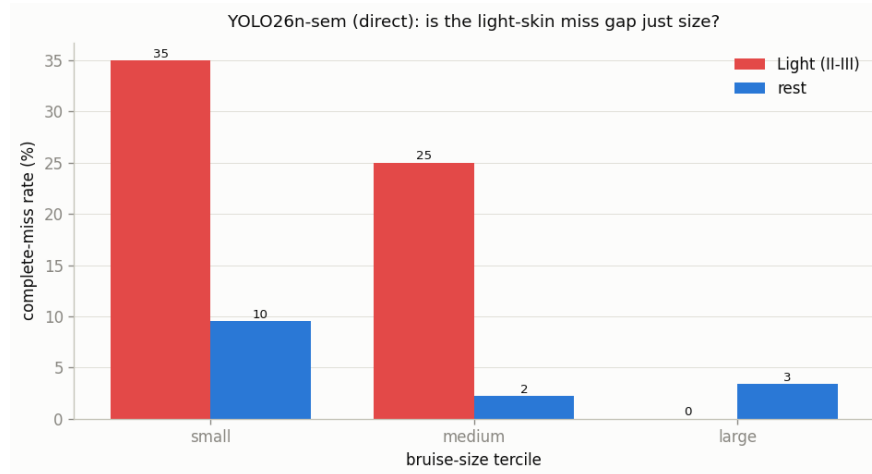


Figure 14: YOLO-direct complete-miss rate, light vs. rest, within each size tercile. The gap does not vanish under size control—it is largest exactly where lesions are smallest.

6 Bruise Size

Small bruises are intrinsically harder (a few pixels of error cost proportionally more Dice), and misses concentrate on them (Figure 15). Recall degrades on the smallest lesions for the YOLO models while SegFormer stays flatter (Figure 16).

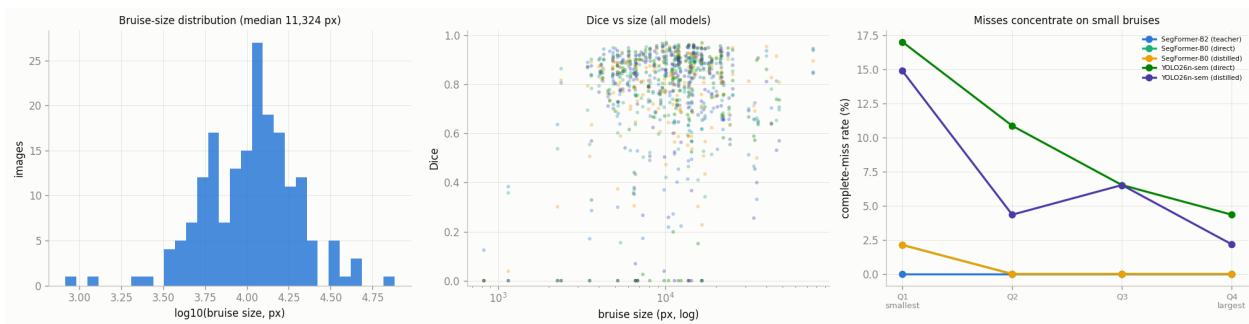


Figure 15: Bruise-size distribution (median 11,324 px), Dice vs. size, and complete-miss rate by size quartile. Misses concentrate on the smallest bruises for the YOLO models.

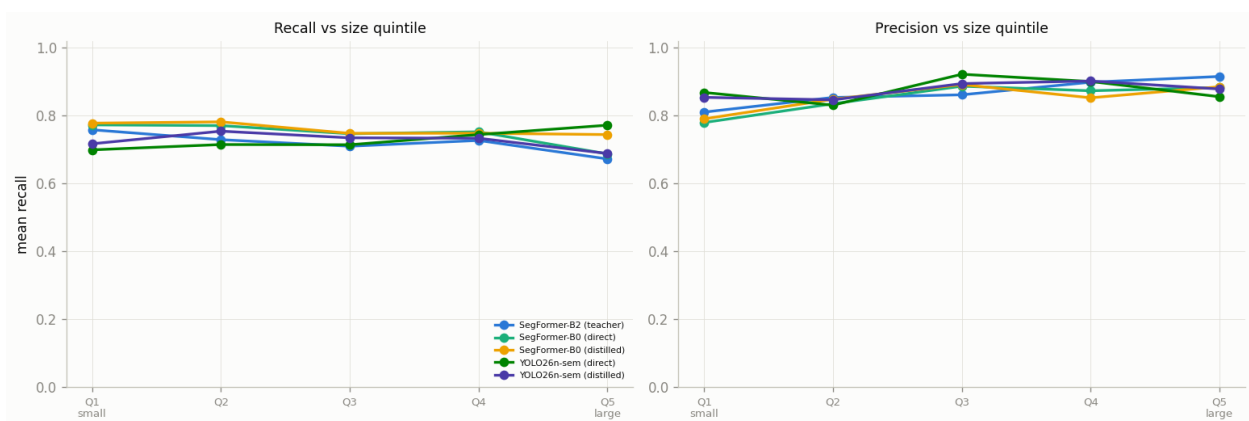


Figure 16: Recall and precision by bruise-size quintile. Precision is roughly flat; recall is where the smallest-lesion penalty shows up.

7 The Annotation Ceiling

The most important result concerns the labels. The three experts agree with one another at only **0.639** mean Dice (Figure 17); every model scores above that against the consensus. Paul—whose masks the models train on—is the outlier, matching the majority least (0.700) of the three.

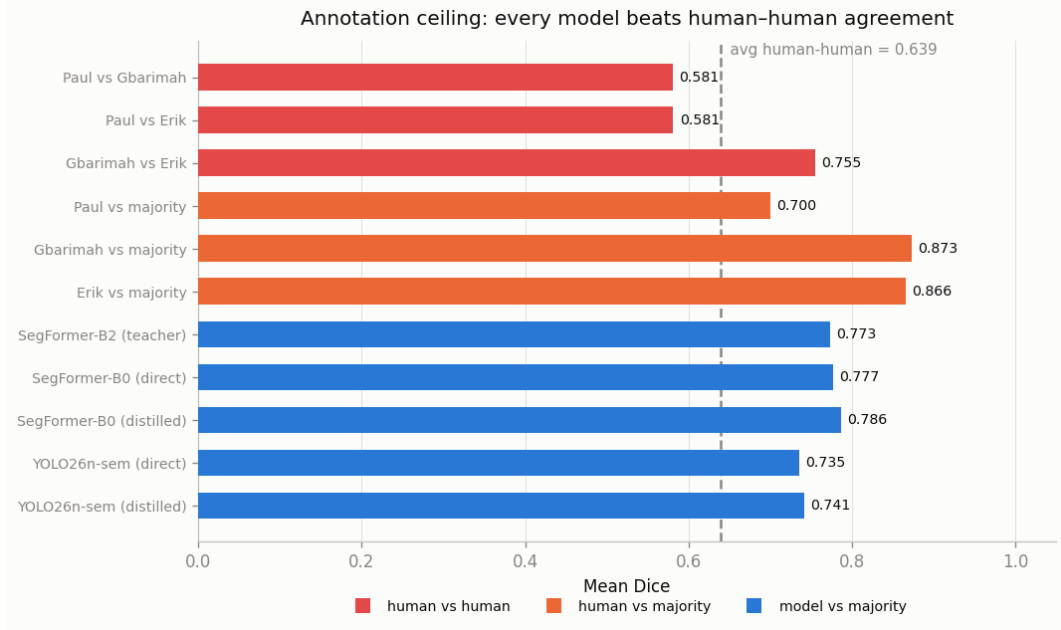


Figure 17: Inter-annotator agreement vs. model-vs-consensus. Every model beats the human-human average (dashed, 0.639).

7.1 The model beats the annotator it learned from

Each model trains only on Paul’s masks yet is scored against the three-expert majority. Table 4 and Figure 18: all three SegFormer models agree with the consensus *significantly better than Paul does*. This effect ($\approx +0.08$) is an order of magnitude larger than any model-vs-model gap, and it is the study’s only significant accuracy result—the models are **label-limited, not capacity-limited**.

Table 4: Model vs. Paul against the same majority target (paired subject bootstrap, $B = 4000$).

Model	Model vs. maj.	Δ (model – Paul)	95% CI	$P(\Delta > 0)$
SegFormer-B2 teacher	0.773	+0.073	[+0.019, +0.131]	0.998
SegFormer-B0 direct	0.777	+0.077	[+0.010, +0.140]	0.988
SegFormer-B0 distilled	0.786	+0.087	[+0.018, +0.156]	0.994
YOLO26n direct (native)	0.735	+0.036	[−0.028, +0.107]	0.85
YOLO26n distilled (native)	0.741	+0.042	[−0.022, +0.107]	0.89

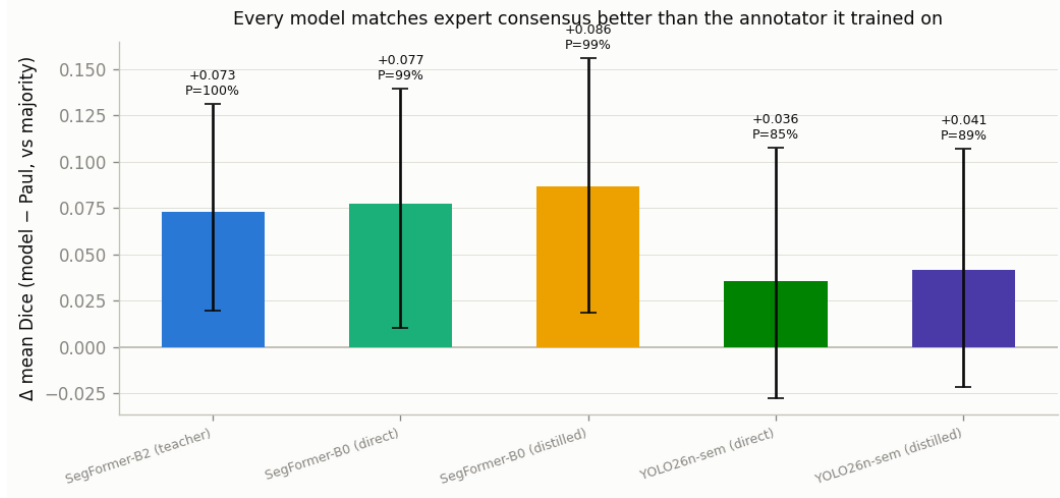


Figure 18: Δ Dice (model – Paul) vs. the same majority, with subject-bootstrap intervals. All three SegFormer bars clear zero.

8 Efficiency

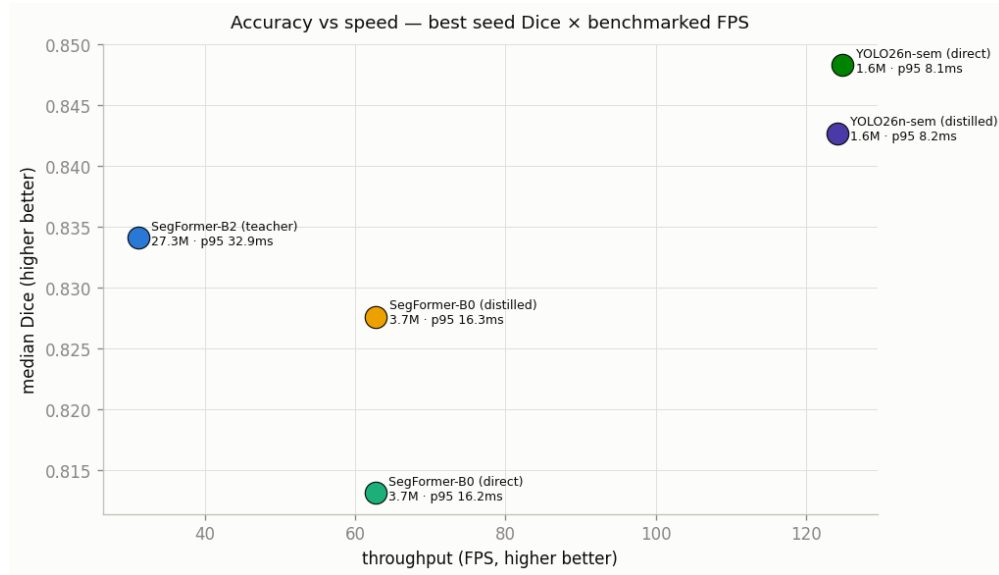


Figure 19: Accuracy vs. speed (median Dice $\times$ benchmarked FPS). YOLO is far faster (~ 125 FPS, 1.6 M params) but pays in miss rate; SegFormer-B0 is the balance point; the B2 teacher is the accuracy anchor at ~ 31 FPS.

9 Qualitative Galleries



Figure 20: Predictions across easy $\rightarrow$ hard images (green = truth, red = model). The hardest row is a tiny, faint bruise that every model but the B2 teacher misses entirely.

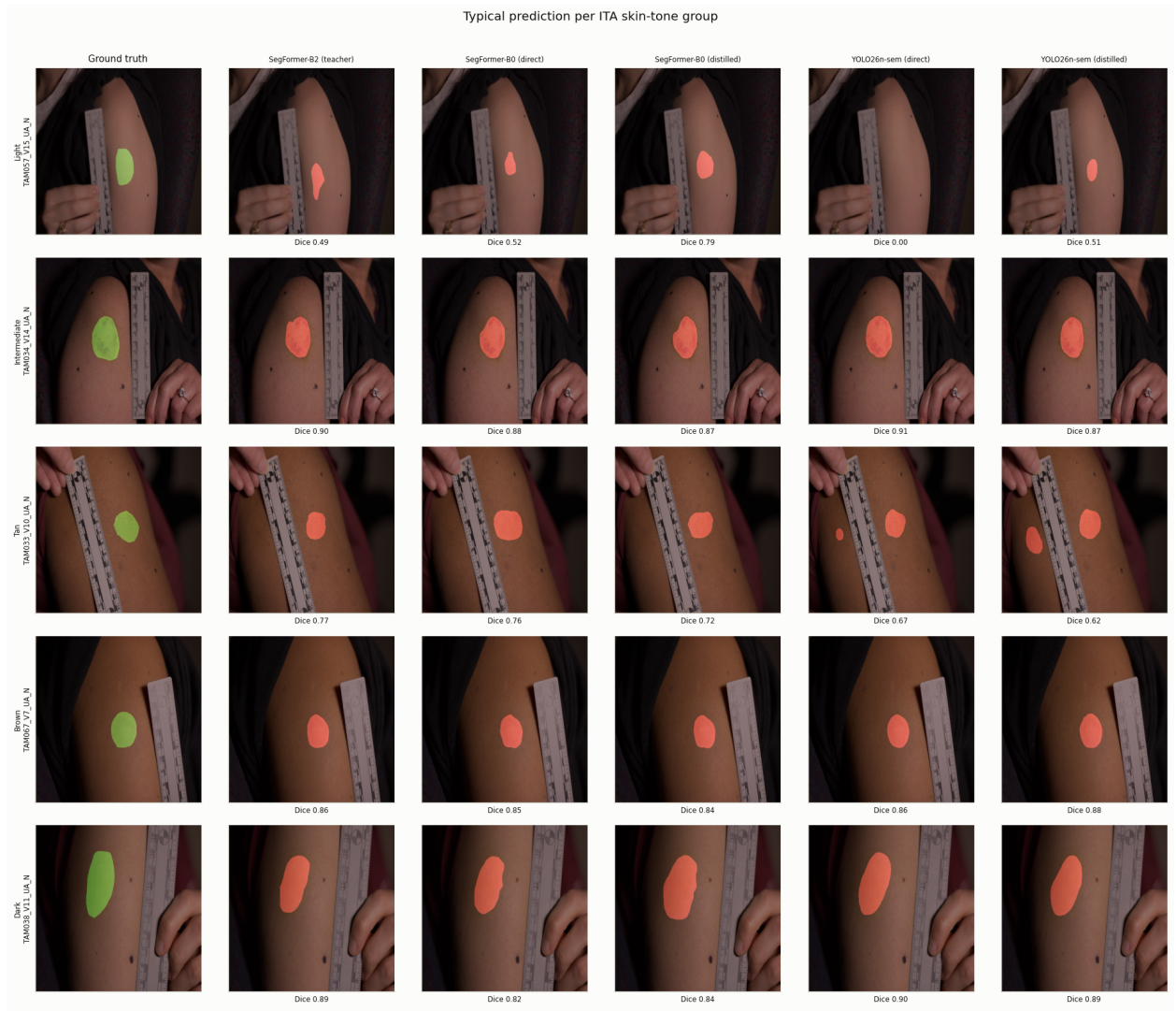


Figure 21: One representative image per ITA skin-tone group. Note the light-skin (top) row, where YOLO-direct returns nothing (Dice 0.00) while SegFormer still localises the bruise.

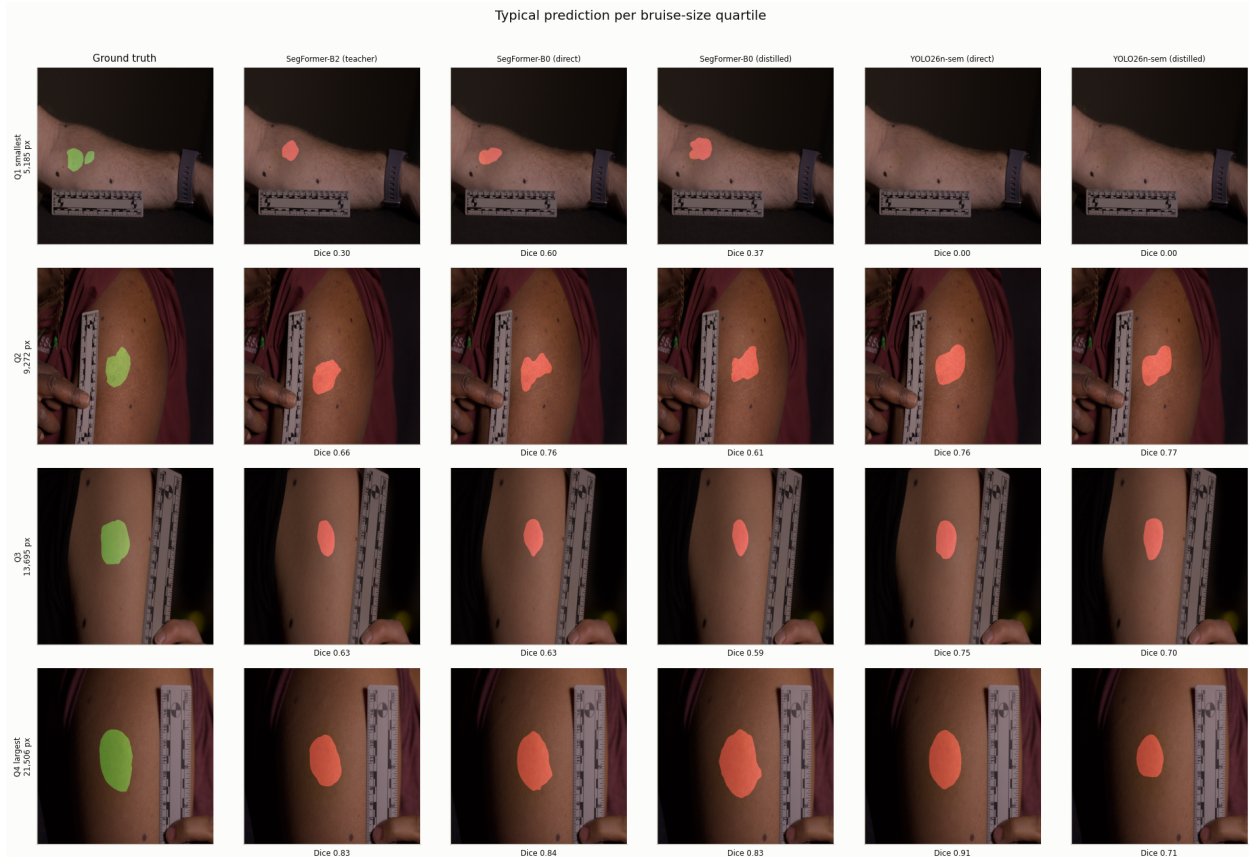


Figure 22: One representative image per bruise-size quartile. The smallest (top) row is where the YOLO models blank; all models handle the largest lesions well.

10 Discussion: how these saved models compare to the retrained run

These are the project’s *original* checkpoints (one per model, no seed replication); the batch-equalised, natively-retrained five-model run is reported separately in `final_run_report.pdf`. The two agree on the structural findings and differ on two points worth stating plainly:

1. **Student vs. teacher.** Here the distilled B0 (0.786) tops the B2 teacher (0.773); in the retrained run, once batch/step budget was equalised, the B2 led. Both gaps are within the annotation noise (Δ n.s.), so neither ordering is a real capability claim—but it is a clean example of how a confound (training budget) can flip a headline that the statistics never supported.
2. **Fairness signal.** On these saved models the residual light-skin recall deficit is significant (after size + clustering) for *most* models including SegFormer, slightly stronger than the retrained run. At 9 light-skin subjects it remains a hypothesis, but a consistent one across both model sets.

Everything else matches: SegFormer is essentially miss-free while YOLO is not; the miss rate, not Dice, is the axis that separates architectures by more than label noise; and the annotation ceiling

(models beat the annotator they learned from) is the dominant, and only significant, accuracy result.

11 Reproducibility

- **Evaluation:** `bruise_colab_saved_eval.ipynb` (generator scripts/46_generate_saved_eval_notebook.py) — loads the saved checkpoints via the `pipeline` package, sweeps the threshold on val, scores test, YOLO two ways, ITA fairness, annotation ceiling. Outputs: the `saved_eval_*` zip.
- **Analysis / figures:** `bruise_colab_saved_analysis.ipynb` (generator scripts/47_generate_saved_analysis.ipynb) — the 22 figures reproduced here, plus the paired contrasts, size-conditioned regression, and galleries. Outputs: the `saved_analysis_*` zip.
- All tabular numbers are read directly from those two zips; the figures are the notebooks' own exports (`analysis/LATEST_IMAGES/BEST_RESULTS/`). Nothing is hand-transcribed.